
Typology Aware mBART-50

Dominik Kysel
Aarhus University
Dept. of Computer Science
au803266@uni.au.dk

Malthe Poulsen Husum
Aarhus University
Dept. of Computer Science
202005586@post.au.dk

Bora Tarlan
Aarhus University
Dept. of Computer Science
au800363@uni.au.dk

Abstract

The project explores a typology-aware multilingual machine translation approach using the SMOL dataset, which contains professionally translated text for 221 low-resource languages. The documentation provides rich information, including script type, language family, and Glottocodes, allowing us to implement language and script family embeddings. The goal is to investigate whether integrating such metadata directly into a multilingual model can improve translation quality for underrepresented languages. Using mBART-50 as the base architecture, we construct “typology embeddings” for each language by adding metadata about script type, a language family and geographical region, and simple corpus-derived indicators such as average sub-tokens per word and type-token ratio. These embeddings are added to the model’s language-ID representations during fine-tuning, enabling the model to learn script-, family- and region-sensitive patterns. Evaluation using chrF and BLEU metrics will determine whether these lightweight, in-dataset typological signals enhance cross-lingual transfer and overall translation performance in low-resource settings.

1 Contributions

This project introduces a typology-aware multilingual translation approach that uses only SMOL’s internal metadata and documented information - script type, language family, and basic linguistic statistics - to improve low-resource translation. The novelty lies in creating data-driven typology embeddings built on SMOL dataset and documentation and integrating them into the mBART-50 model’s language-ID representations. This method enables the model to adapt to each language’s structural traits without external databases, offering a reproducible way to enhance cross-lingual transfer in multilingual MT.

2 Methods

The initial work on the project was a review of the SMOL dataset and its published paper to understand the construction, language coverage, and the make-up of its metadata. This review was followed by a screening of related research using SMOL, identifying preprocessing strategies that were commonly employed and current limitations in low-resource multilingual translation. This review of the academic landscape gave us intuitions on how we intend to structure and make use of SMOL’s metadata fields, such as script type from the ISO 15924 column, continent-based family indicators, and Glottocodes.

An exploratory data analysis assisted us in characterizing SMOL’s linguistic and structural diversity. From this we derived languages which had enough data in documents and sentences, while ensuring diversity in family and region. By doing this we initially decided on a set of 18 languages, however after further consideration we narrowed it down to 15.

Next, the data preprocessing step ensured consistency and allow for seamless model integration. This step includes the cleaning and the normalization of text, applying mBART-50’s SentencePiece model for subword tokenization, and encoding each language with its proposed metadata fields. For each language, a typology embedding were constructed by concatenating a one-hot encoding of script type, an embedding of family/region which will be learned, and the continuous morphological indicators. These embeddings will be combined with mBART-50’s existing vectors, allowing for the model to consider both structural and typological information in machine translation.

Fine-tuning conducted on SMOL’s multilingual parallel data using mBART-50 as the base model. The training objective mirrored standard sequence-to-sequence translation fine-tuning. However, the difference was at both encoder and decoder stages, where the introduction of typology-enhanced embeddings served as auxiliary inputs. Experiments compared baseline mBART-50 performance with and without typology embeddings to observe their contribution.

In the evaluation step, metrics such as chrF and BLEU were utilized to assess translation quality across multiple low-resource directions. We conducted both aggregate and per-language analyses to measure improvements and understand which typological factors help the model learn better connections between languages.

3 Related work

SMOL is proposed as training data to aid machine translation efforts into the so-called low resource languages in the article that introduces the dataset (Caswell et al., 2025). Low resource languages inherently bring about challenges in machine translation objectives due to having little to no publicly available sources available. The paper reports significant improvements in translation quality based on prominent evaluation metrics when used for fine tuning or prompting LLM’s.

Multilingual language models that are normally regarded as capable of performing reasonably well in cross-lingual settings of NLP tasks have been evaluated in higher resource settings. Alongside this observation, the importance of pretraining techniques and additional data was highlighted for low resource language translations (Wu and Dredze, 2020).

It has been demonstrated before that overcoming the difficulties of translations into under-resourced languages can be achieved without the help of joint training alongside higher resource languages. Ogueji et al. (2021) has shown that a model trained on more focused yet smaller training data can outperform pretrained multilingual language models.

Work done by Oncevay et al. (2020) forms an example of language family relatedness being used as a feature to improve multilingual machine translation. Their method that derives typological features and language phylogenies of source languages manages to encapsulate language relationships, which then in turn yields significant improvements in neural machine translation tasks.

4 Implementation

4.1 Dataset

To properly test our hypothesis we need a dataset that is both small enough to be manageable for training and iteration, as well as large enough to include different script types, regions and language families in enough variation to allow an embedding to be learned. However, the SMOL dataset is not as expansive as we would have liked. Not every language contained enough data to be considered a candidate. For instance, some languages only contained data in the Gatitos space, meaning 4000 single word translations, which is not conducive to training since they will invariably be unable to predict the single-word translations of words they’ve never seen.

We therefore decided on choosing only languages with SmolSent entries, containing 900 sentences in the source and target language. With this restriction, we looked at the amount of languages remaining and categorized them based on shared region, script type, and family relations, provided they shared that category with at least 2 other language entries. This resulted in 3 regions, 2 script types, and a number of language families, as SMOL is a relatively Africa-centric dataset, where most languages are in latin script.

Because the only three languages from Asia also shared a language family and script type, each embedding would theoretically be the same. Moreover, since there is no native tokenizer for devanagari in mBART-50 which made the output extremely poor, and because they had significantly less data than other languages (Containing no SmolDocs), we decided that they would not contribute properly to the dataset. After cutting the Asia region, Deva script and Indo-Aryan family in one go, just 1 script type (Latin), 2 regions (Europe and Africa), and 5 language families remained:

Table 1: Dataset

Germanic	Italic	Central Mande	Lwoo	East Bantu
Spanish(EU)	Icelandic(EU)	Bambara(AF)	Acholi(AF)	Swahili(AF)
Ligurian(EU)	Nigerian Pidgin(AF)	Dyula(AF)	Alur(AF)	Tswana(AF)
Mauritian Creole(AF)	Krio(AF)	Susu(AF)	Luo(AF)	Bemba(AF)

The languages were chosen to ensure that they were as close as possible geographically within their language family, and with enough data to see if certain language families would perform better than others.

Finally, we choose to work primarily with SmolSent and SmolDocs, ignoring Gatitos. The thinking was that since we are trying to find out if the model can learn shared characteristics of a language, having the syntax and morphological structures present would be an advantage, and it also drastically cut the computation time, allowing more flexibility for iterating with limited computation power.

The dataset lands at a training size of 32114 training sentences, 1290 validation sentences, and 2640 test sentences, spread out over 15 different languages.

4.2 Training, evaluation and hyperparameters

To train the model, we finetuned the existing mBART-50 a dataset comprised of SmolSent and SmolDocs entries for our chosen languages. We chose a 70/20/10 data split, however, because there were wildly varying amounts of data available between languages, we chose to standardize so that the amount of testing and validation entries were the same across the board. This allowed more training entries for languages with more resources, and it sped up the evaluation process significantly, while almost always giving better results.

For evaluation metrics, we chose BLEU and chrF. BLEU measures the N-gram overlap of words between reference and prediction, while chrF measures the N-gram overlap between characters.

For training, we used a variety of hyperparameters. Notably, we chose a batch size of 8 due to memory constraints while training, and a reasonably low learning rate of $3e-5$ so that mBART-50 would not lose too much of its innate language understanding, as it was already capable of certain languages in the dataset (Spanish) or related languages (French, which is close to Nigerian Pidgin). We added warmup steps to ensure the model would have time to adjust slowly to these new languages, again in an effort to preserve the translation capabilities of mBART-50.

Finally, it’s important to note that we used the SentencePiece tokenizer that accompanies mBART-50 to tokenize source and target texts. This invariably results in some level of inconsistency in languages with unique morphology, but for our purposes, it’s sufficient, as we will be comparing languages to themselves, not each other.

4.3 Custom typology implementation

To allow the model to learn features from region, script and language family, the mBART-50 model was wrapped in a custom model, augmenting the original forward, initialization and generate functions.

The initialization function extends the base mBART-50 model with additional components for typological conditioning. The model introduces three learnable embedding tables that encode script, language family, and geographical region as discrete categorical features. Each typological feature is embedded in a reduced-dimensional space with the objective of limiting parameter growth. The embeddings are then concatenated and passed through a linear projection layer that maps them back

to mBART-50’s hidden dimension. This ensures that the typological information is integrated while the internal architecture of mBART-50 remains unaltered.

The forward function augments the standard mBART-50 training pass so that the model receives typology identifiers along with tokenized inputs and converts them into embeddings. These embeddings are concatenated, projected and broadcasted to create a vector which is added to the encoder’s token embeddings before being processed by the transformer layers. The modified embeddings are passed to the mBART-50 model, which has remained internally unchanged, so the integrity of the attention mechanisms, decoding and loss computation is maintained. This means that the typological information is incorporated into all encoder layers during training.

The generate function mirrors the typology conditioning applied in forward pass. This is to ensure that the resulting vector, which is constructed as described above, is consistent with the training process. This vector is then incorporated into the encoder token embeddings, after which these embeddings are passed to the mBART-50’s native generate method.

5 Results

We trained the model in various configurations. Notably, we trained it with 1, 3 or 5 epochs, and trained it either with the standard mBART-50, both typology embeddings, or just the family embeddings. For the 1 epoch runs, we trained the model from scratch 5 times using 5 different partitions of the dataset, to try and average the results and dampen the effects of randomness (Table 2, Table 3):

Table 2: BLEU averages

1 epoch, BLEU			
Language	Baseline	Region, and Family Δ	Family only Δ
Spanish	27.86	+ 0.66	+ 0.40
Ligurian	11.28	+ 0.51	+ 0.32
Mauritian Creole	10.54	+ 0.64	+ 0.99
Icelandic	0.41	+ 0.05	- 0.08
Nigerian Pidgin	39.24	+ 0.77	+ 0.41
Krio	15.63	- 0.07	+ 0.06
Bambara	6.66	+ 0.47	+ 0.39
Dyula	2.92	+ 0.33	+ 0.16
Susu	1.40	+ 0.31	+ 0.11
Acholi	2.57	+ 0.22	+ 0.03
Alur	1.60	+ 0.65	+ 0.36
Luo Pidgin	5.21	+ 0.04	+ 0.12
Swahili	18.04	+ 0.52	+ 0.45
Tswana	1.83	+ 0.09	+ 0.04
Bemba	2.78	- 0.22	- 0.28

Additionally, we trained the model with 5 epochs to see if the results were more pronounced (Table 4, Table 5). Due to computing constraints, we were not able to do more than once per configuration, and therefore this data should not be considered final:

While other considerations were tried, they mostly led to similar results but with higher compute times. Increases in epochs caused slight increases in delta scores across the board, but large gains in base model performance, as expected.

6 Results analysis

The results showed a general increase across the majority of languages, based on the average values produced by the three model types. While it is a small increase on average, it is important to note that this is in contrast to generally small values.

chrF showed indications of better and more consistent scores in models that had trained for more epochs. It may be that word order n-grams, which BLEU is a measurement of, simply is inconsistent when dealing with languages that are more closely- or more distantly related.

Table 3: chrF averages

1 epoch, chrF			
Language	Baseline	Region, and Family Δ	Family only Δ
Spanish	55.14	+ 0.52	+ 0.60
Ligurian	37.04	+ 0.66	+ 0.29
Mauritian Creole	37.06	+ 0.41	+ 0.49
Icelandic	12.07	+ 0.62	- 0.91
Nigerian Pidgin	64.41	+ 0.40	+ 0.38
Krio	35.47	+ 0.30	+ 0.47
Bambara	18.09	+ 1.32	+ 1.19
Dyula	14.05	+ 1.12	+ 1.16
Susu	9.64	+ 0.45	+ 0.35
Acholi	16.25	+ 1.00	+ 0.43
Alur	12.08	+ 2.04	+ 1.25
Luo	24.89	+ 0.30	+ 0.50
Swahili	43.84	+ 0.69	+ 0.60
Tswana	15.37	+ 0.25	- 0.06
Bemba	18.27	- 0.06	- 0.27

Table 4: BLEU scores

BLEU			
Language	Baseline	Region, and Family Δ	Family only Δ
Spanish	20.66	+ 4.38	+ 2.36
Ligurian	22.90	- 0.40	+ 0.69
Mauritian Creole	28.00	- 1.17	+ 0.54
Icelandic	2.46	-0.84	+ 0.19
Nigerian Pidgin	45.13	+ 2.76	+ 2.13
Krio	29.78	+ 0.81	+ 1.31
Bambara	20.05	+ 0.80	+ 0.65
Dyula	10.11	+ 0.82	+ 0.07
Susu	5.58	- 0.85	+ 0.50
Acholi	8.17	- 1.16	+ 0.33
Alur	7.97	- 0.08	+ 1.17
Luo	14.93	-1.52	+ 0.17
Swahili	34.64	+ 1.32	+ 3.17
Tswana	11.68	+ 1.62	+ 1.44
Bemba	7.31	- 1.37	+ 0.16

Table 5: chrF scores

chrF			
Language	Baseline	Region, and Family Δ	Family only Δ
Spanish	50.12	+2.15	+ 1.16
Ligurian	46.98	- 0.53	+ 1.11
Mauritian Creole	52.29	+0.22	+ 1.37
Icelandic	19.60	- 0.01	- 0.81
Nigerian Pidgin	68.58	+ 1.42	+ 1.23
Krio	49.18	+ 1.76	+ 1.55
Bambara	39.17	+ 1.72	+ 1.55
Dyula	30.52	+ 1.00	+ 0.36
Susu	21.58	+ 0.28	+ 1.05
Acholi	28.90	- 0.03	+ 1.16
Alur	28.07	+ 1.36	+ 1.82
Luo	40.91	- 0.69	+ 0.54
Swahili	61.30	+ 0.64	+ 1.58
Tswana	33.81	+ 1.27	+ 0.99
Bemba	32.30	+ 1.19	+ 0.97

Including script type and region impacted some languages more than others. Icelandic, for instance, seems to have benefited from a shared embedding with other European languages, which is unsurpris-

ing, as it is very distantly related to the other germanic languages geographically. But overall, the differences were small. In higher epoch training, we have indications that regions add noise in the outcome, and excluding it was more stable.

- While the results are not as significant as we had hoped, we do find that having consistent small increases across almost every language is a step in the right direction. There are many reasons why the results may not have been as significant as we had hoped.
- We may simply not have had the compute to train models as much as they needed for a more distinct increase in scores to become apparent. We trained a model with up to 5 epochs, which gave better and much clearer results, but we were not able to average results with models trained on other training partitions due to compute limitations, and this therefore remains speculation.
- Another issue may be training size variance for the languages. Some languages, like Swahili, had a lot more training data available, leading to better baseline results. But this also might cause the embeddings to learn more from Swahili, neglecting the other languages from its family / region.
- It is also possible that some languages are simply more distantly related to those in its family / region. For instance, Icelandic, which not only had a small dataset, but is both distantly related to the English-based contact languages it shares a family with, and distantly related to the other European languages, may not have been able to benefit much, if at all, from these embeddings. Geographic and cultural similarity may have to be prioritized more for a smaller, more focused dataset in the future.
- Finally, we believe that packing morphology and syntax of a language family or region into a single embedding is a large task, and a single embedding alone may not be able to represent all the facets of the language that its grouped languages need to improve their translation. It may therefore be important to explore stronger alternatives to a single tensor.

7 Conclusion

Our study focused on improving the performance of a multilingual machine translation model, mBART-50, on low resource languages. The literature has demonstrated that even with small amount of data available, multilingual neural translation models achieve relatively impressive results on these underrepresented languages. To that end, in our approach, the SMOL dataset is employed, and a set of typological features are added to the mBART-50 architecture. The experimental setup compares three settings of mBART-50 with and without the addition of our novel features.

While the results are not conclusive, they are promising, indicating that a refined methodology and higher available compute may result in more significant improvements to translation quality, both in BLEU and chrF, but more refining is needed.

Another limitation in our workflow that can be pointed at is the use of mBART-50's tokenizer. By nature, this model has a single tokenizer that is able to handle its native languages with which it was trained. However, by using it for the new low resource languages we introduce to the model, we are assuming that this tokenizer will perform reasonably well for these languages, enough so that we can measure an improvement in translation. This assumption is contested by previous work that has demonstrated the potential improvements of employing monolingual tokenizers for multilingual models, therefore emphasizing the effect of having a specialized tokenizer for the languages (Rust et al., 2021).

Our study sets a precedent as an effort to improve neural machine translation into low resource languages by combining the focused dataset provided by SMOL with additional typological embeddings derived from the language metadata. We believe that further research in novel feature engineering methods to aid machine translations may be valuable in overcoming the difficulty in improving results of NLP tasks regarding under-resourced languages.

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